

Note 6.2 Volumes Using Cylindrical Shells

Slicing with Cylinders

- **Strategy:**

We can slice with cylinders straight down through the solid so that the axis of each cylinder is parallel to the y -axis. Then unroll each cylindrical shell, whose volume is the product of the area $A(x)$ of the rectangular slab and the thickness Δx .

- **Example:**

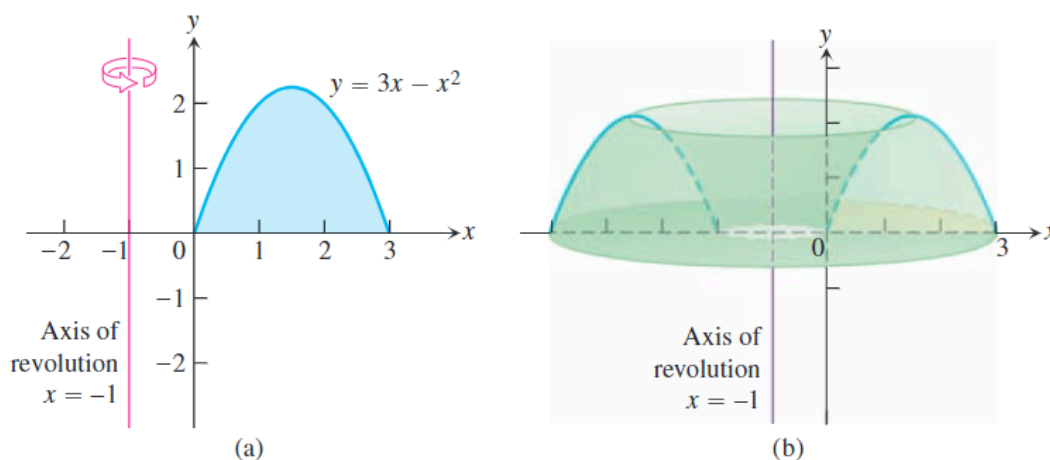


FIGURE 6.16 (a) The graph of the region in Example 1, before revolution.
(b) The solid formed when the region in part (a) is revolved about the axis of revolution $x = -1$.

For this example, it is awkward to use the washer method since we need to express the x -value in terms of y and its formula is complicated. So we can slice the solid into cylindrical shell.

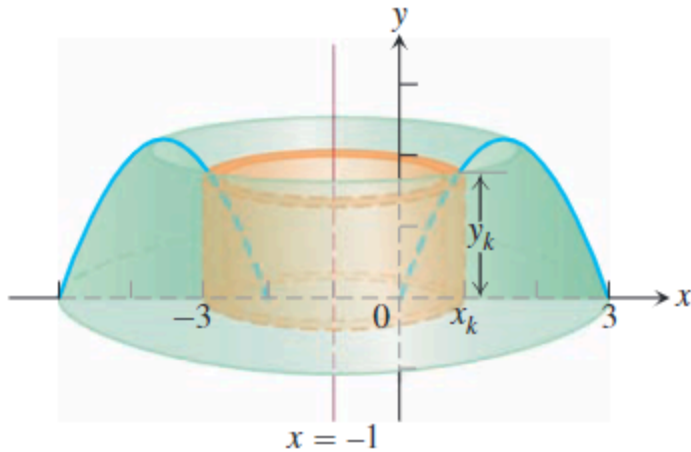


FIGURE 6.17 A cylindrical shell of height y_k obtained by rotating a vertical strip of thickness Δx_k about the line $x = -1$. The outer radius of the cylinder occurs at x_k , where the height of the parabola is $y_k = 3x_k - x_k^2$ (Example 1).

then each slice is setting over a subinterval of x -axis of width Δx_k so that the thickness of the cylindrical shell is Δx_k . And its radius is $1 + x_k$ so that the length of it is $2\pi(1 + x_k)$. Then the height is $y(x_k) = 3x_k - x_k^2$.

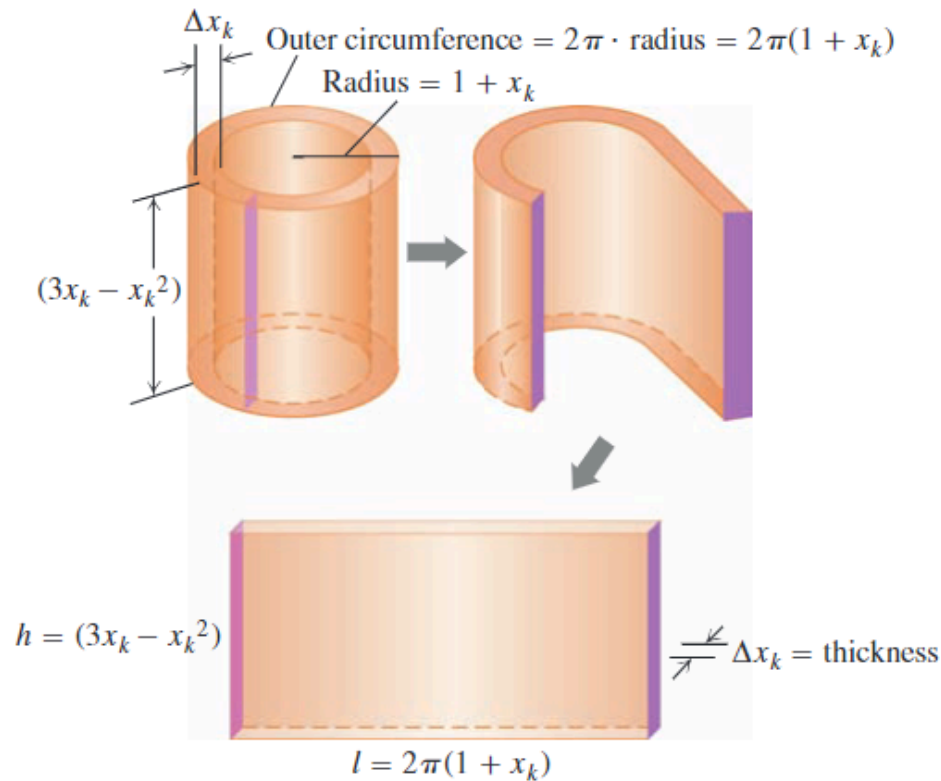


FIGURE 6.18 Cutting and unrolling a cylindrical shell gives a nearly rectangular solid (Example 1).

Then the volume of every cylindrical shell is

$$\Delta V_k = 2\pi(1 + x_k) \cdot (3x_k - x_k^2) \cdot \Delta x_k$$

So the sum of individual shells gives the Riemann sum

$$\sum_{k=1}^n \Delta V_k = \sum_{k=1}^n 2\pi(1 + x_k) \cdot (3x_k - x_k^2) \cdot \Delta x_k$$

When $n \rightarrow \infty$, $\|P\| \rightarrow 0$, that is, $\Delta x_k \rightarrow 0$,

$$V = \lim_{n \rightarrow \infty} \sum_{k=1}^n 2\pi(1 + x_k) \cdot (3x_k - x_k^2) \cdot \Delta x_k = \int_0^3 2\pi(x + 1)(3x - x^2) dx$$

The Shell Method

- Shell Formula for Revolution About a Vertical Line**

The volume of the solid generated by revolving the region between the x -axis and the graph of a continuous function $y = f(x) \geq 0$, $L \leq a \leq x \leq b$, about a vertical line $x = L$ is

$$V = \int_a^b 2\pi(x - L)f(x) \, dx$$